

Canada Environmental Science Opportunity

Applicant Guide for Stream 1: Research,
Modelling & Risk Assessment and Stream
2: Nature Research



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Environment and Climate Change Canada
Public Information Centre
Place Vincent Massey building
351 St-Joseph Boulevard
Gatineau, Quebec K1A 0H3
Toll free: 1-800-668-6767
Email: enviroinfo@ec.gc.ca

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What is the Canada Environmental Science Opportunity?

Environment and Climate Change Canada (ECCC) is launching a new funding opportunity for environmental science.

The Canada Environmental Science Opportunity (CESO) seeks to engage experts from across Canada to advance projects that address critical environmental issues, such as climate change and nature as aligned with [ECCC's mandate](#).

For projects initiating in fiscal year 2026-2027, applicants are welcome to apply to the following streams:

- Stream 1: Research, modelling & risk assessment
- Stream 2: Nature research
- Stream 3: Research & analysis to support Canada's path to net zero
 - Please note that Stream 3 of the CESO is delivered through the Environmental Damages Fund (EDF). For more information, refer to the [Environmental Damages Fund webpage](#) – particularly the [Available Funds section](#).

The CESO uses an open call delivery model to foster transparency and fairness and leverage capacities and synergies within the Canadian science community.

Prioritizing Canadian

The Government of Canada is taking steps to protect and prioritize Canadian workers and industries and strengthen Canada's domestic economy. Applicants are encouraged to prioritize Canadian suppliers, materials and content in carrying out their project.

What is the purpose of this guide?

This Applicant Guide provides an overview of the funding opportunity and instructions for submitting an application to the CESO call for applications, including how to apply through the online Grants and Contributions Enterprise Management System (GCEMS).

Applications must include all required information outlined below and meet the eligibility criteria to be considered for funding.

Who can apply?

This call is limited to Canadian applicants. Detailed eligibility requirements for each stream and theme are provided in the relevant sections of this guide.

Applicants may work independently or in collaboration with other eligible organizations, including academic institutions, not-for-profit organizations, Indigenous partners, and private sector organizations.

While multiple partners may collaborate to develop an application, the applicant responsible for submitting the application via GCEMS will be considered the lead applicant. If selected for funding, the lead applicant will be responsible for negotiating and signing a funding agreement with ECCC. They will be responsible for the ongoing delivery of the project and for reporting on results and will be accountable for the use of the funding.

Applicants are eligible to apply to multiple streams and themes under CESO as long as they meet the eligibility criteria. Please note that a separate application must be submitted for each stream and theme in which you wish to be considered.

Applicants must provide an identification number for their organization in their application, such as a Charitable Number, Non-Profit Organization Registration Number or First Nations Band Number.

ECCC encourages applicants to consider equity, diversity, and inclusion in the design, management, and implementation of their projects.

Please note that applicants will be required, in their project application, to attest to how their project and its partners align with the [Policy on Sensitive Technology Research and Affiliations of Concern](#) and that they are not a member of, or affiliated, with any organization on the list of [Named Research Organizations](#).

What projects may be funded?

Projects that are eligible for funding under the CESO must enable activities that will contribute to Canada's environmental science capacity in priority areas, such as climate change and nature. Projects are also encouraged to facilitate the transfer of knowledge and integration of that knowledge such that it can inform the further advancement of environmental science.

Eligible activities vary by theme. While general activities such as research, capacity building, and networking are common across all themes, certain themes may include additional priorities such as climate mitigation, biodiversity conservation, ecosystem restoration, or weather and climate services. Applicants should consult the theme-specific guidelines for detailed eligible activities.

Stream 1: Research, modelling, & risk assessment

Projects must align with **one or more** of the following five research themes:

Theme 1A: Research related to Canada's changing climate

Contribution Funding

Up to \$150,000 per year, per project for a maximum of four years.

Eligible applicants

Canadian post-secondary academic institutions, and Canadian regional climate consortia.

Science overview

Projects funded under this theme will address key knowledge gaps in climate science, where evidence remains limited or uncertain.

(1) Carbon cycle feedbacks over land and/or oceans

Research will reduce uncertainty in the magnitude, sign, timing, and regional variability in carbon cycle feedbacks using improved modelling methods which may include Artificial Intelligence (AI)/Machine Learning (ML) methods.

Priority

Funded projects will focus on one or more of the following priority science and knowledge areas:

- Advancing process understanding and modeling of carbon feedbacks over land and oceans, and how these feedbacks affect the global natural carbon budget or Canada's net carbon balance
- Quantifying uncertainties in carbon cycle feedbacks to improve long-term projections

(2) Regional climate change, including extreme events

Research will address low confidence aspects of regional climate signals and extremes. This includes event attribution, high impact or compound extremes, and high-resolution regional climate modelling.

Priority

Funded projects will advance one or more of the following priority science and knowledge areas:

- Improving attribution and projection of high-impact extreme events
- Advancing high-resolution regional modelling of extremes
- Large-scale atmospheric drivers of extremes including changes in low frequency variability modes (e.g. El Niño-Southern Oscillation – ENSO, Pacific Decadal Oscillation - PDO, North Atlantic Oscillation - NAO), storm tracks, the polar vortex, extratropical cyclones, atmospheric rivers, and North Atlantic hurricanes
- Improving historical characterization and understanding of poorly observed extremes, including sub-daily rainfall, freezing rain, hail, and convective wind events

(3) Probabilistic scenarios of future change

Research will support the development of probabilistic, decision-relevant scenarios that identify plausible futures and communicate uncertainty transparently.

Priority

Funded projects will advance one or more of the following priority science areas:

- Providing probabilistic projections for temperature, precipitation, and related risk indicators aligned with the needs of adaptation practitioners
- Developing methods to reconcile multi-model ensembles, observational constraints, and storyline approaches to reduce divergence and quantify uncertainty in future regional change

(4) Extreme Coastal waves and water level projections

Research will provide new insights into coastal water level projections under different climate scenarios, including extreme events.

Priority

Funded projects will advance one or more of the following priority science and knowledge areas:

- Providing probabilistic projections of future changes in coastal water levels
- Applying coastal downscaling of ocean wave conditions to hundreds of meters, including information of the wave spectra, needed for coastal management and other applications
- Developing and applying wave setup and wave runup projections (a component of the extreme water level along a shoreline not currently available in projections at the Canadian scale)

- Advancing attribution studies based on coastal waves and/or sea level data (e.g., for coastal flooding)
- Determining sea ice effects on coastal waves and water levels, including improvements in sea ice parameterizations in wave models based on recent satellite data

Theme 1B: Quantifying the impact of cryosphere loss on freshwater

Contribution Funding

Up to \$200,000 per year, per project for a maximum of five years.

Eligible applicants

Canadian post-secondary academic institutions.

Science overview

The cryosphere is sensitive to climate change. Across Canada, snow, glaciers, permafrost, lake ice, river ice, and sea ice have changed markedly in recent decades. Glacier decline, permafrost thaw, and shifts in snowmelt timing affect freshwater quantity and quality. Funded projects will expand knowledge of processes and feedbacks, modernize monitoring systems, and apply advanced numerical modelling to understand cryosphere responses and implications for aquatic ecosystems and society. Science that advances understanding of cryospheric change will support water resource management and climate adaptation in Canada and contributes to the UN Decade for Action on Cryospheric Sciences (2025–2034).

Priority

Funded projects will advance one or more of the following priority science and knowledge areas:

- Improving observational and modelling systems to quantify and predict how thawing permafrost affects surface and groundwater quantity and quality, including releases of metals, carbon, nutrients, and contaminants
- Quantifying how warming, reduced snow, and declining glacier runoff affect freshwater storage and supply, with implications for agriculture, municipal water, hydro-electricity, and groundwater recharge
- Studying links between cryosphere change, freshwater habitats, and aquatic ecology, including effects of changing snow and glacier meltwater on stream temperatures
- Projecting shorter lake and river ice seasons and assessing effects on shoreline erosion, water temperature, evaporation, water levels, water quality, aquatic ecosystems, ice roads, and weather (e.g., lake-effect snow)

- Understanding regime shifts in freshwater quantity, quality, and aquatic ecosystems related to temperature, precipitation, and cryosphere change. This work will guide risk assessment and support the development of integrated hydro-climatic monitoring and modelling initiatives

Theme 1C: Studying changing air quality conditions with climate change

Contribution Funding

Up to \$100,000 per year, per project for a maximum of four years.

Eligible applicants

Canadian post-secondary academic institutions, Canadian non-governmental organizations, and Indigenous governments, boards, commissions, associations, authorities.

Science overview

Unlike other forms of pollution, every person living in Canada is exposed to air pollution, which presents the largest environmental risk to human health in Canada, contributing to an estimated 17,400 premature deaths and \$146 billion per year in economic costs. Air quality and climate change interact, with some air pollutants also contributing to warming or cooling the climate, so researchers must consider these links. Higher temperatures, extreme weather, and other climate-related changes are expected to worsen air quality.

Wildfires are occurring more often and with greater intensity, compounding climate change's impacts on air quality. Smoke (particulate matter, or PM) is the most visible issue, but wildfires also release large amounts of other toxic pollutants, and the fuel that feeds the fire determines the mix. From 2013–2018, wildfire smoke contributed to an estimated 620–2,700 deaths in Canada. The public health burden from wildfire smoke is likely to increase.

Additionally, the processes of PM and ozone formation are changing as the mix of emitted pollutants changes. Canada has achieved significant reductions in emissions of the smog precursors nitrogen oxides (NO₂), sulphur dioxide (SO₂) and volatile organic compounds (VOCs). Despite decreases of key precursors, ambient concentrations of PM and ozone have not decreased significantly at the national level in the past 10 years.

Funded projects will examine the interdependency of air quality and climate change to quantify ecosystem and human health impacts. They will use these findings to inform emissions risk management strategies.

Priority

Funded projects will advance one or more of the following priority science and knowledge areas:

- Understanding how the changing mix of emissions impacts atmospheric chemistry, air quality, and pollutant deposition
- Quantifying wildfire emissions and studying how they affect air quality, using historical data from events where wildfires affected communities and ecosystems
- Understanding how urban emission profiles will change with climate change, and how those shifts will affect air quality, including how researchers will consider the transport of wildfire smoke into urban areas
- Examining how transportation emissions could evolve, and the implications for air quality
- Identifying the pollutants whose emissions are expected to increase significantly and thus should be tracked in the future

Theme 1D: Building confidence in the use of new approach methods in ecological risk assessment

Contribution Funding

Up to \$250,000 per year, per project for a maximum of four years.

Eligible applicants

Canadian post-secondary academic institutions.

Science overview

Research is needed to develop, standardize and establish scientific relevance and confidence to demonstrate the suitability of new approach methods (NAMs) in keeping with the Organisation for Economic Co-operation and Development ([OECD](#)) [weight of evidence guiding principles and key elements](#). This is to support fit-for-purpose uses of NAMs across regulatory contexts. Research that advances efforts to develop NAMs or integrated approaches to testing and assessment (IATA) that could eventually replace existing acute and chronic fish assays in ecological risk assessment is of particular interest. In addition, research that advances the development of omics-based points of departure is of high interest. Consideration of tiered approaches and the [OECD guidance on the generation, reporting and use of research data for risk assessment](#) are recommended in study design. The results of the funded research projects will strengthen capacity in Canada to prevent and manage pollution and to implement the [Strategy to replace, reduce or refine vertebrate animal testing under the Canadian Environmental Protection Act, 1999 \(CEPA\) - Canada.ca](#).

Priority

Funded projects will aim to build confidence in the use of NAMs in ecological risk assessment, and advance one or more of the following priority science and knowledge areas:

- Computational modelling
- Biochemical assays
- Cellular / *in vitro* assays
- Early-life stage (that is, prior to self-feeding) and non-vertebrate whole organism models
- Tissue and organoid models
- Data science / bioinformatics

Theme 1E: Advancing the integration of Artificial Intelligence (AI) for operational hydro-meteorological and environmental services

Contribution Funding

Maximum of \$200,000 per project for one, two or three years, with funding for multi-year projects allocated in equal amounts each year.

Eligible applicants

Canadian post-secondary academic institutions actively engaged in research related to the atmosphere, environment, water, air quality, information technology, or data science.

Science overview

Projects will advance expertise in the integration of Artificial Intelligence (AI) technologies for operational hydro-meteorological and environmental services in Canada.

- Innovations in AI technologies for weather and environmental prediction including modelling, post-processing, data management, and observation quality control have the potential to deliver broad societal benefits, such as improved, timely and user-specific information related to extreme and high-impact weather and environmental events, more efficient use of computing resources, and enhanced responsiveness to emerging and evolving service needs
- Projects are sought to advance knowledge and expertise in Canada to better understand how AI can be effectively applied in the context of operational hydro-meteorological and environmental services, while remaining aligned with principles of transparency, accountability, and responsible use of AI

Priority

Funded projects will advance one or more of the following areas:

- AI-based methods that improve quality of predictions for extreme and high-impact weather and environmental events.
- Unified AI-based approaches and frameworks across atmospheric, oceanic, hydro, land and air quality domains, with an emphasis on improving modelling performance and computational efficiency
- AI-based decision-making and prioritization activities in weather and environmental services, using large data sets
- AI-based methods to assess, enhance and maximize the value and reliability of observational networks
- AI-based intelligent water monitoring, prediction, and resource optimization

Stream 2: Nature research

Theme 2A: Nature-based climate solutions

Contribution Funding

Up to \$100,000 per year, per project for a maximum of three years.

Eligible applicants

Canadian post-secondary academic institutions, Canadian non-governmental organizations, and Indigenous organizations, governments, boards, commissions, associations, authorities.

Science overview

Addressing climate change is an urgent need that requires avoidance and reduction of greenhouse gas emissions (GHGs), mitigation of impacts, and adaptation to change. However, some potential actions that could contribute to these objectives can have unintended negative consequences for nature and biodiversity. Projects will enable the reductions of Canada's net GHG emissions using natural climate solutions, while benefitting biodiversity and human well-being. Projects will address science and knowledge gaps to be better able to effectively design and implement nature-based initiatives to avoid or reduce GHG emissions and mitigate climate change impacts while benefitting biodiversity.

Priority

Funded projects will advance one or more of the following priority science and knowledge areas:

- Avoid GHG emissions by halting or reducing the conversion of carbon-rich ecosystems to less carbon-rich ecosystems (e.g., from a forest/grassland/wetland to cropland or an urban development)
- Restore carbon storage and biodiversity in converted or degraded ecosystems
- Reduce GHG emissions caused by natural resource use, including by implementing new forest harvesting management practices
- Develop policies, programs or tools with key actors in priority sectors, and implement them to reduce GHG emissions and improve biodiversity
- Measure and report on GHG outcomes from natural climate solutions
- Provide biodiversity and human well-being co-benefits
- Advance the federal commitment to reconciliation with Indigenous people
- Advance science and knowledge on natural climate solutions in Canada

Theme 2B: Species at risk

Contribution Funding

Up to \$100,000 per year, per project for a maximum of three years.

Eligible applicants

Canadian post-secondary academic institutions, Canadian not-for-profit organizations, and Indigenous organizations, governments, boards, commissions, associations, authorities.

Science overview

There are more than 600 species at risk (SAR) listed under [schedule 1](#) of the [Species at Risk Act](#). For many other species, we lack information to properly assess their population status. Projects will enable species-at-risk-focused scientific research on migratory birds, other migratory species, and species that span multiple Canadian jurisdictions.

Priority

Funded projects will advance one or more of the following priority science and knowledge areas:

- Informs the assessment of [COSEWIC candidate species](#), including drivers of their population decline
- Informs [recovery planning for species listed as Threatened or Endangered](#), including [resident description](#) and [critical habitat identification](#) for which a recovery strategy has not yet been prepared ([view all existing reports](#))

- Addresses the description of activities detailed in the schedule of studies in existing [recovery plans](#)
- Inform [stewardship and conservation activities](#), including the assessment of effectiveness/appropriateness of recovery actions for [SAR species](#)

Theme 2C: Building the science to support the protection of coastal ecosystems

Contribution Funding

Up to \$100,000 per year, per project for a maximum of three years.

Eligible applicants

Canadian post-secondary academic institutions, Canadian non-governmental organizations, and Indigenous organizations, governments, boards, commissions, associations, authorities.

Science overview

Projects will advance research on the stressors impacting marine birds and other wildlife in arctic ecosystems, and community-based scientific training and capacity-building programs that foster marine wildlife and ecosystem science expertise in coastal communities in Canada. This science will contribute to ensuring healthy, resilient, and productive coastal ecosystems.

Priority

Funded projects will advance one or more of the following priority science and knowledge areas:

- Research to inform the assessment of risks of marine transportation to marine birds and other wildlife, particularly in the Arctic
- Research to develop species distribution models for Arctic marine species and other wildlife
- For community-based project components, funding will support developing and delivering science-training programs on marine ecosystems, especially marine birds and other wildlife. These programs will enable Indigenous peoples to engage in conservation and in assessing impacts within their lands or traditional territories

What projects and activities will *not* be funded?

The following types of projects and activities are *not* eligible to receive funding from the CESO:

- Activities required by law and/or mandated by any level of government

- Containment and clean-up of environmental spills
- Restoration of contaminated sites
- Infrastructure¹, particularly related to municipal, provincial, and federal government program areas
- Lobbying or advocacy activities
- Recreation and tourism projects or beautification initiatives
- Preparation of formal curriculum materials²
- Core organization functions and activities such as meetings, maintenance, and administration (however, project specific administrative support is eligible)
- Annual or regular organization events/campaigns
- Expenses to attend general conferences and workshops (project specific conferences and workshops are eligible)
- Activities already underway
- Projects outside of Canada
- Land acquisition
- Bursaries or cash incentives

What funding is available?

- ECCC funding is available for up to 100% of eligible project costs; matching funds are not required, but other sources of funding (cash or in-kind) are considered an asset
- Applicants must declare all potential funding sources, including federal, provincial, territorial, municipal, and other contributors
- Eligible projects must build capacity and support activities that strengthen domestic environmental science capacity
- Applications from networks of academics are encouraged and should strengthen science capacity across Canada; contributors and partners should facilitate knowledge transfer and integration of that knowledge such that it can inform the further advancement of environmental science
- Eligible projects and activities are described in the “What projects may be funded” section and under each stream/theme of this guide

¹ Expenses related to assembly of infrastructure (i.e., including labor costs and equipment rental) are ineligible. However, a project could use funding sources outside of CESO funding to fund assembly of the infrastructure, and utilize CESO funding for the following expenses:

- Dismantling of existing infrastructure in a state of disrepair
- Removal of the dismantled materials
- Disposal of the old materials
- Planning of new infrastructure (i.e., engineering contracts for planning purposes, legal expenses)
- Acquisition of new materials
- Transport of materials related to the infrastructure

² Formal curriculum materials are defined by being a planned program of objectives, content, learning experiences, resources and assessment offered by provincially/territorially funded schools, or privately funded schools.

What is the maximum duration of a CESO project?

The maximum duration for a CESO project is dependent on the stream and theme.

When is the application deadline?

The deadline for submitting a funding application through the GCEMS portal is **15:00:00 (3:00 pm) Eastern Time**, on the date indicated in [GCEMS](#).

How to apply

Step 1

[Log into GCEMS](#) using your GCKey to access the application. You can create a new GCKey by following the onscreen prompts.

Thoroughly review all information and criteria on the Funding Opportunity page when accessing the GCEMS application form, as well as in this Applicant Guide, to ensure the project satisfies all requirements. If the project does not meet all required eligibility criteria, it will not be eligible for funding. For questions or clarification, please contact the CESO inbox (OSEC-CESO@ec.gc.ca).

Should you be unable to submit your application using [GCEMS](#), please contact the CESO inbox (OSEC-CESO@ec.gc.ca). Please note we aim to respond to all enquiries within three (3) to five (5) business days, depending on volume.

For any technical issues related to website functionality and access (e.g., GCKey and SWIM), GCEMS technical support can be reached at sgesc-gcems@ec.gc.ca.

Step 2

Develop your CESO funding application using the GCEMS online portal and visit the [GCEMS application instructions page](#) for technical assistance documents, tutorials, and support throughout your application preparation.

As the website will automatically log applicants out after 15 minutes of inactivity, you are strongly advised to save your draft application in GCEMS frequently or to copy and paste content from an external document to avoid losing your work.

If desired, contact the CESO inbox (OSEC-CESO@ec.gc.ca) prior to the application deadline to discuss the application process with a CESO Program Officer. Officers can also help provide advice/information on:

- The CESO program and eligibility
- The funding process

- Official languages requirements

Please note the CESO Program Officers can only provide guidance if it pertains to the above-listed information. To ensure transparency and fairness for all applicants, they will not provide general input or feedback on potential projects.

A strong CESO application:

- Demonstrates that the applicant and proposed activities meet eligibility requirements and clearly align with the CESO, the relevant stream/theme, [ECCC's mandate](#), and any relevant supporting documentation, where applicable
- Clearly describes the environmental issue(s) or knowledge gap(s) being addressed, the project need (local, regional, or national context), and the expected environmental benefits
- Presents a technically feasible project with a realistic timeline, clear work plan, measurable objectives, and defined expected results
- Demonstrates organizational capacity, relevant expertise (or access to it), experience managing similar projects, sound project management, and appropriate risk identification and mitigation (including health, safety, and regulatory considerations)
- Includes a clear monitoring, evaluation, and performance measurement approach
- Provides a detailed and reasonable budget that represents good value for money
- Demonstrates confirmed or potential cash and/or in-kind contributions and meaningful engagement with affected communities, Indigenous groups, partners, or consortia (letters of support where applicable)
- Leverages ECCC data, expertise, or tools where appropriate and considers adoption of the Open Government License – Canada
- Optional to include Contributors³ or Partners⁴, but including them can be beneficial for demonstrating collaboration and knowledge-sharing potential

³ Contributors are organizations that will receive a portion of the funding contributed to the Lead Applicant, participate in implementing project activities, and contribute to the overall success of the project. To be eligible for funding, Contributors must fall within one of the eligibility categories identified above for Lead Applicant. Contractors and consultants hired to carry out project activities are not considered Contributors.

⁴ Partners are organizations that will not receive a portion of the funds contributed to the Lead Applicant but will support the project's success through financial and/or in-kind contributions. Partners may include organizations that are otherwise ineligible to receive funding, such as for-profit organizations, associations, and municipal, provincial/territorial, or federal governments.

Step 3

Your application must be submitted before the stated deadline in GCEMS by **15:00:00 Eastern Time**. The CESO does not accept applications via email, mail, or fax. Once you have submitted your application, you will receive an email from GCEMS acknowledging the receipt of your submission. **If you do not receive confirmation**, please contact sgesc-gcems@ec.gc.ca.

Following the project review phase, you will receive a notification on the funding decision of your application. If you do not receive confirmation, please contact OSEC-CESO@ec.gc.ca.

When contacting ECCC about an application, always reference the GCEMS application ID number associated with the application.

Guide to completing an online application

The following information will enable you to complete an online project application for the CESO. Note, the CESO Applicant Guide is not a technical guide for navigating GCEMS but rather provides CESO program-specific information to ensure project applications meet the CESO program eligibility. For any questions regarding navigating GCEMS, please refer to the GCEMS Technical Assistance for the Creation of a GCKey Guide on the [Environmental funding programs: Application page](#).

The following modules must be completed. Please follow the instructions provided in each section.

Applicant information

The Applicant Information module is a summary of organizational and contact information that auto-populates various modules of your GCEMS application. You must complete all fields:

- Organization legal name, address, email, phone number, and preferred official language of communication
- Contact information for the principal applicant representative (the individual representing the project and responsible for, and knowledgeable about, the project and application)
- Select your organization type, provide an organization identification number (for example, charitable number, non-profit organization registration number or First Nations band number), and provide a brief description of your organization, including its mandate
- Identify and describe previous funding received from ECCC and/or other Canadian federal government departments or agencies
- Describe any major transformations your organization has undergone in the last two years (for example, changes to the board of directors, major restructuring, or hiring of key personnel)

- Disclose any funds your organization currently owes to the Government of Canada. This may include overpayments from previous agreements or other amounts due. Note that owing funds does not prevent your organization from applying for or receiving funding.

Project information

This module provides an overview of your project. All information is mandatory.

Project title: A short, descriptive title to identify your project.

Anticipated project start date/end date: When determining the proposed project start date, consider that expenses are eligible once the approval notification is issued. As the exact timing of decisions may vary, applicants should choose a project start date that allows for flexibility.

Project location: Provide your project's location; please note that the project must be undertaken within Canada. If your project has multiple activity sites or is national in scope, note this in your application in the Project Summary field and provide the coordinates for your organization's head office or primary office location in Project Location.

Project objectives: Clearly state the objective(s) of your project and how it relates to the CESO and the relevant stream/theme. Provide a clear description of specific and measurable objectives that are achievable and realistic during the timeframe of the project. Your project objectives should clearly link to the funding you are applying for.

Project summary: Provide a summary description of your project from start to finish, showing the link between the project and the CESO and the relevant stream/theme.

It is important that your summary be concise and includes the key result(s) (performance indicator(s) you expect to achieve with your project.

Is there any other relevant information concerning your project we should know?: Include any other information about your project that you feel is important to know. This could include further details on project site(s) area or size, nearby watercourses or areas of importance, restricted access, etc. If your project requires permits, licenses, or special authorization, complete the question on Permits and Authorizations later in the application.

Project team experience: Describe relevant qualifications and experiences of your project team members that could speak to the ability to carry out the proposed project. Clearly explain how their skills, knowledge, and past work relate to the project objectives. Highlight specific expertise that supports successful delivery (e.g., technical skills, field experience, community engagement, cultural knowledge, scientific or restoration expertise, etc.). Include any information on project partners in this section.

Financial capacity: Describe your organization's financial capacity to manage this project. This could include outlining other sources of funding for your project as detailed in the budget module, such as letters of support to indicate support from partnering organization(s).

Additionally, this could include a summary of past successful projects, demonstrating your organization's experience managing a project budget of similar size and scale.

Project management capacity: Describe your organization's experience in managing and delivering projects of this size, scale, and scope. Provide details on the duties and responsibilities of the position(s) identified in the human resources section of the budget. Include who will work on the project (e.g., project coordinator, volunteers, consultants, technical experts, etc.) and their roles/qualifications. Identify relevant qualifications and experiences (e.g., ability to track finances; effectively plan, execute and monitor projects, including defining objectives, schedules and allocating resources; etc.) of the project team. This section is an opportunity to demonstrate your organization's experience and ability to carry out the proposed project.

Contractors: The use of contractors is an eligible expense under the CESO. If you plan to use contractors, describe what role they will have and who the contract will be with. These details do not need to be finalized when you apply, but include any information known at time of application.

Capital assets⁵: Capital assets are an eligible expense under the CESO, though projects that focus primarily on the purchase or installation of capital assets or equipment are not eligible. If your project includes capital assets as a planned or anticipated expense, describe what items will be purchased and how they will benefit the project.

Project work plan

Complete the information to explain the main activities that will be completed throughout the project. Indicate a title for each activity.

Include a description of the activity, which must include site-specific details as relevant to the project. Provide details of the main tasks to be completed as part of each activity. You must include information such as timelines, techniques, and/or methods. You may consider organizing activities by type or into fiscal quarters according to the following breakdown: April to June, July to September, October to December, and January to March.

Each main activity must include outputs, outcomes, and expected results. Outputs are the tangible "products" of the activity such as reports, data sets, and outreach materials. Outcomes are the intended change or benefit resulting from the outputs, such as a desired change in behaviour; for example, the area of fish habitat restored, or the number of hectares restored. Expected results are specific, measurable indicators tied directly to the activity and should outline how achievement will be demonstrated.

Your work plan must include communications activities and/or tasks relevant to your project where appropriate and how relevant partners and/or stakeholders will be informed of and/or

⁵ Capital assets are defined as tangible assets costing more than \$10,000 with a useful life of more than one year.

invited to participate in the project. Tasks may be related to (but are not limited to) bringing awareness to the project or a project activity and announcing the project or a project activity's results. Communications outputs may include publicly available reports, presentations, posters, and news releases. Outcomes and expected results may include the number of people reached or the number of communications outputs that will be created and delivered.

The work plan should clearly link your activities to the budget in your application. As applicable, the work plan should also clearly demonstrate the consultation, engagement, or participation of priority groups as described elsewhere in your application.

Evaluation plan and performance measures

Expected results: Present the expected results of your proposed project (i.e., the overall impact of your project) based on the project's objectives. Ensure that the expected results are achievable based on the project activities presented in the work plan.

Project evaluation plan: Describe how you will assess your project's overall success (e.g., site visits, surveys of target audiences, analysis of project results, etc.). It is important to clearly demonstrate how overall project results will be captured.

Identify a specific evaluation plan to measure the progress and success of your project and the methodology that will be used to capture results. For each Key Performance Indicator (KPI) included in the next section, describe the methodology used to estimate and/or calculate the indicator value.

Key performance indicators: Performance indicators are measurable results that demonstrate how your project objectives will be met. Target values should represent outcomes for the entirety of your project. Ensure that your target project indicators are achievable within the timeframe of CESO funding. Include the name, title, and/or description of your indicator; the target (numerical) value; and the unit to measure the target value (e.g., number of species targeted).

CESO - Research - Advancing the integration of AI for operational hydro-meteorological and environmental services - 2026/2027

Please note that the program-specific module is specific to only **Theme 1E: Advancing the integration of Artificial Intelligence (AI) for operational hydro-meteorological and environmental services**. It does not apply to any other themes.

Please complete the modules in the order they appear in your application. The system will automatically display the correct module for your stream.

The following questions must be completed:

1. Please identify if projects will involve students or post-doctoral researchers (yes or no).
 - a. If yes, mandatory second and third questions will appear.

2. Please describe your plans for involving students or post-doctoral researchers and specify the capacity in which they will participate.
3. Please list the titles of all known project researchers on the project team and indicate which individuals are students and/or post-doctoral researchers. Do not include any personal information.

Project budget

This module will present your project's financial details. Complete all three steps to prepare a budget using the tables provided in the online GCEMS application. Numbers should be rounded to the nearest dollar.

Step 1: Enter contributors

Requested funding: Clearly state the total amount of CESO funding requested for each year of your project. Note: The CESO does not provide in-kind services and should not be listed as an in-kind supporter.

Contributors for project: Add all other contributors to the project (including your own organization's contribution to the project). Provide a Contributor name, identify a Contributor Type, and whether the contribution is confirmed at the time of submitting your application. Other optional contributor types may include support from other federal government departments, provincial governments, municipal governments, Indigenous organizations, or other partners.

Consider the following elements:

- Volunteer time is valuable; calculate all volunteer time at fair market value and include it in the budget as an in-kind contribution
- Expected revenue generated by the project must be included in the budget; provide an approximate value of the expected revenue, which is expected to be used toward project activities
- Do not include land value in matching funds
- Present a balanced budget
- If indicating ECCC in-kind participation (e.g., professional advice, laboratory work, or analysis), provide letters from the supporting organization confirming their involvement and, where possible, include the estimated value of any in-kind or cash contributions; these documents can be uploaded in the "Other Supporting Information" section of the application

Step 2: Enter expenditures

Expenditure table: Add all project expenses to the table. For each expense, select an expenditure type from the list and provide a brief but thorough description of the details or further breakdown of the expense. Include the dollar value that will be covered by CESO funding

(in the **Requested ECCC funding** total cash column) or other project contributors as applicable. The **What project costs are eligible?** section provides further details on eligible expenditures for the CESO.

Step 3: Review and edit contributors and expenditures

Summary of contributors and expenditures: Your contributors and expenses must match to submit your application. This table will indicate if your budget is not balanced. You can adjust your contributors and expenditures in the following table or return to step one and two.

Review and edit budget information: Use the drop-down menu to display the contributors and expenditures by each fiscal year of your project and use the tabs to switch between the two. In this table you can add new contributors and expenses or edit existing entries.

Tips for completing a strong budget:

- Separate CESO funds from other ECCC and federal funds, if applicable
- Avoid using vague wording in the budget by providing enough detail to link budgeted costs with project activities
- Ensure the budget per fiscal year matches the project work plan per fiscal year
- Include sufficient detail or a breakdown of expenses in the expenditure description field so that it is clear how you arrived at the dollar value
- Ensure the budget per fiscal year is sufficient to reach the objectives and indicators per fiscal year
- Ensure the evaluation of income and expenditures is accurate, and the financial contribution request is well-justified and appropriate
- All expenditure estimates should be at fair market value; fair market value represents the standard rate for a product or service in your area
- If project partners have been identified in the 'Project Team Experience' section, and they will be making a financial contribution (including in-kind contributions), ensure that this information is included in the project budget

What project costs are eligible?

Costs related to the following types of expenses are eligible under the CESO program:

- **Salaries and wages**
 - Include salaries and benefits of employees who are directly working on the project (for example biologists, project coordinators, technicians, bookkeepers, accounting)
 - If bundling all salary and wages into a single expenditure item, include details (for example, number of positions supported) of the bundled costs in the expenditure description
- **Management and professional service expenditures**

- Management and professional service costs (e.g., accounting, project monitoring, translation); translation services can include Canada's official languages as well as other languages required to effectively reach and support [linguistic minority communities](#)
- **Contractors**
 - Include engineering, construction, research, or other relevant activities
 - Expenditures included under contractors should align with the information provided in the Applicant Information module
- **Travel**
 - Include only travel that is directly related to carrying-out project activities
 - Include mileage and other costs required for the achievement of fieldwork.
 - Refer to the [National Joint Council's Appendix B](#) for your province/territory kilometric rates, and the [National Joint Council's Appendix C](#) for lodging, per diem, or other related items
- **Materials and supplies expenditures**
 - Include only items required for the achievement of project activities such as office supplies and tools and equipment (including field, lab, and sampling equipment)
- **Equipment rentals**
 - Include only equipment needed for the project; rental of vehicles has its own cost category and must not be included here
- **Vehicle rental and operation expenditures**
 - Includes expenses such as insurance and fuel
 - To determine eligible costs and the amount ECCC deems reasonable, refer to the "Transportation" section of the [National Joint Travel Directive](#) and the [Public Service and Procurement Canada \(PSPC\) Accommodation and Car Rental Directory](#)
- **Printing, production, and distribution expenditures**
 - Costs associated with project-specific communication and outreach materials (for example, brochures or fact sheets)
- **Further disbursement of ECCC funding to an ultimate recipient⁶(s)**
 - Funds disbursed to an ultimate recipient by means of the applicant's own contribution-type program must respect CESO terms and conditions, including CESO eligibility requirements
- **A reasonable share (20% or less) of overhead and/or administrative expenditures directly related to carrying out the project**
 - Indirect costs necessary to support the achievement of the overall project objectives, that cannot be linked to a specific activity or are not significant enough to detail under a specific cost category (for example, insurance utilities, repairs and maintenance expenses, memberships and subscriptions, IT support, bank charges, etcetera)
- **Costs associated with an Indigenous participation or ceremony**

⁶ An ultimate recipient is an individual or entity that will receive a payment from the initial recipient. The ultimate recipient only has a relationship with the initial recipient; they do not have a relationship with the department.

- Monetary honoraria offered to Indigenous Elders and/or Knowledge Keepers related to specific participation activities
- Includes costs related to translation and interpretation, leading a traditional opening or closing ceremony or prayer, sharing of traditional knowledge and protocols, and demonstrating traditional art and practices
- **Hospitality expenditures**
 - Expenditures must be in accordance with the [Treasury Board Secretariat's Directive on Travel, Hospitality, Conference, and Event Expenditures](#)
- **The purchase of eligible carbon offset credits for events, conferences, and travel**
 - Vendor must have received third party verification and adheres to internationally agreed standards (including Gold Standard, VER+, VERRA, B.C. Offset System)
- **Expenditures for preparing an independently verified financial audit**
 - Projects expecting to spend more than \$100,000 in a single fiscal year should include this cost in the final year of their budget
- **Purchase of capital assets⁷** (note that projects focused primarily on the purchase or installation of capital assets or equipment are not eligible)
 - Tangible assets costing more than \$10,000 with a useful life of more than one year
 - Expenditures included under capital assets should align with the information provided in the Applicant Information module
- **Other expenditures**
 - GST/HST reimbursement

Please note, this is not a complete list. You can contact the CESO inbox (OSEC-CESO@ec.gc.ca) for any matter relating to this list or the eligibility of your project and activities.

Stacking limit

The stacking limit for this opportunity is 100% of eligible project costs. This means total government funding (federal, provincial, territorial, municipal) for the same expenses cannot exceed the project's total eligible costs. Applicants must declare all other funding sources. ECCC's contribution will cover only what is needed to achieve the project's goals.

What expenditures are *ineligible*?

- Lobbyist fees
- Costs related to production, sale, or export of softwood lumber
- Land securement
- Costs related to preparing or submitting the project application

⁷ If, during the CESO agreement, it becomes necessary to acquire capital assets, the ownership of such assets purchased in whole, or in part, with federal money may, if agreed to, rest with the Recipient. Acknowledgment of the federal funding shall be indicated in such fashion as the affixing of a federal identity mark in accordance with the federal identity program

Independently verified financial audit

Should your application be successful, details regarding financial reporting requirements (including potential completion of an Independently Verified Financial Report at the end of the project) will be outlined during the negotiation of the funding agreement.

The costs related to an Independently Verified Financial Report are eligible expenditures if deemed (by ECCC) a requirement for the funding agreement during the negotiation process. If not deemed a requirement, the estimated amounts for an Independently Verified Financial Report can be reallocated to other eligible project costs.

If you expect to spend more than \$100,000 in any one fiscal year, it is recommended to include an estimate of the cost for an Independently Verified Financial Report in the application's budget module, under the final year of the project. This audit cannot be conducted by your organization and must be completed by an external contractor.

Department audits

ECCC reserves the right to audit any CESO project at the Department's own expense. Projects of all scope may be subject to an audit at any time. An audit could occur while projects are currently active, or after projects have ended.

Official languages

Completing the Official Languages module of the application is a mandatory requirement for all applications to ECCC funding programs. Applying organizations must respond to all questions by selecting "yes" or "no".

Depending on your project's audience, and in accordance with the *Official Languages Act*, you may be required to offer products and/or services to the public in both official languages. Ensure any associated costs (e.g., translation) pertaining to project delivery and/or promotion are included in the budget.

Other supporting information

Providing the following supporting information is recommended. Should it not be provided at the time of submission, it may be required at a later date:

- Dated and signed letters from other funding sources confirming cash and in-kind
- Contributions, including the specific dollar value of the cash and/or in-kind contribution being provided
- Dated and signed letters proving that the applicant has engaged with individuals, organizations, and/or communities who may be directly or indirectly impacted by the proposed project (for example, Indigenous groups and organizations)

- Dated and signed supporting letters from the appropriate land authority (e.g., private landowners, First Nation Governments, treaty organizations, or Crown land managers) confirming permission to carry out restoration activities on the land and indicating whether they are providing any monetary or in-kind support.
- Letters of support endorsing the project; these letters are different from partner letters confirming their monetary and/or in-kind contribution
- Any information essential to understanding your project, such as pictures of the environmental damage, or maps of the area
- Proof of incorporation, charity status, current filings with corporate registry, and/or current list of board members, as applicable
- Documentation demonstrating organizational capacity, such as resumes or CVs of key staff, the organization structure, or year-end reports
- Any additional information that will enhance your application (for example, pictures or sketches)
- Copies of permits or approvals, species lists, and other relevant documents, if applicable

Letters that confirm other funding (both cash and in-kind) will be requested from successful applicants before a funding agreement can be signed and funds provided.

Note: Any information submitted after the deadline **will not** be considered in the application evaluation process.

Certification

Applicants must certify that the information stated in the application is complete and accurate. Once your application is complete, submit it using GCEMS.

Please note that the CESO **will not** accept project applications by email, fax, or mail. However, if you experience barriers using the Government of Canada's online application platform (GCEMS) due to a disability, please contact the program (OSEC-CESO@ec.gc.ca) to request an accommodation. Alternative formats or support may be provided to ensure you can complete and submit your application.

What happens after you submit a project application?

Acknowledgment of receipt

Acknowledgment of receipt is automatically generated by GCEMS. **If you did not receive an acknowledgment of receipt, you should contact GCEMS at sgesc-gcems@ec.gc.ca to confirm that your application was received.**

Review

Once the Call for Applications is closed, your application is reviewed by CESO teams to verify eligibility. All project applications are subject to the same evaluation process and criteria and are reviewed to ensure that they provide sufficient information to adequately assess the application and that they meet eligibility requirements.

Following this administrative assessment, your project application is examined by a team of technical reviewers, to evaluate your application's scientific and technical merit. The last step includes conducting a final evaluation of each of the applications received.

If your application is approved in principle, you will be contacted to negotiate a funding agreement that outlines the terms and conditions of the funding. Federal Members of Parliament and/or their team will be advised about the approval in principle of a project and be provided with the following information: applicant name, project title, project description, project location, funding amounts, communication activities, and contact information.

If your application is not approved, you will also be informed of this decision via email.

Any questions or comments may be directed to the CESO inbox (OSEC-CESO@ec.gc.ca).

Contact us

Please note, enquiries will be addressed in the order they are received. We aim to respond to all enquiries within three (3) to five (5) business days, depending on the volume and complexity of the enquiries received.

National CESO Office

E-mail: OSEC-CESO@ec.gc.ca